

Case Study 2: Data Science at Golden State Warriors

By

Jade Winebright

September 18, 2025

“On my honor as an Aggie, I have neither given nor received unauthorized aid on this academic work.”

This is entirely my own work or—if authorized by my instructor—the work of the team I was assigned to, except as documented in a bibliography or acknowledged below.

Signature * Jade Winebright

Artificial Intelligence (AI) Acknowledgement:

Microsoft Copilot assisted with minor re-wording and editing of this report.

Submitted to fulfill the requirement for Case Study 2
DAEN 400, Fall 2025
Department of Industrial & Systems Engineering
Texas A&M University

Executive Summary

The Golden State Warriors' Ticket Timing Model, developed by Ray Yocke, Senior Director of Business Analytics and Strategy, predicts when fans will purchase tickets to optimize promotional email timing. This study aimed to improve the current model. To do this, different machine learning models were tested and compared against each other, ultimately leading to an improved method of targeted emails. These included K-Nearest Neighbors, Logistic Regression, and Random Forest Classification. The Random Forest Classification model performed well, with a validation accuracy of 0.82, and a perfect test accuracy on the five held-out samples (1.00). Additionally, it was concluded that a model can be used to estimate the number of days before a game that a customer will purchase a ticket. Random Forest Regression had a validation R^2 of 0.610 and a test R^2 of 0.901, reducing the mean squared error from 20.06 (multiple linear regression) to 11.18, outperforming The Ticket Timing Model. These results demonstrate that Random Forest models significantly outperform the original linear models and should be used to inform future email campaigns, ultimately increasing ticket sales.

Table of Contents

Executive Summary	ii
Table of Contents	iii
List of Figures	iv
List of Tables	iv
1. Introduction	1
1.1 Project Scope	1
2. Methods	1
2.1 Given Analysis	1
2.2 Exploratory Data Analysis	1
2.3 Machine Learning Models for Predictions	1
2.3.1 Models to Predict Fan Type	1
2.3.2 Models to Predict Number of Days	2
3. Results	2
3.1 Classifying Fan Type	2
3.2 Predicting Days Before Game	4
4. Conclusions	6
4.1 Feature Relationships	6
4.2 Fan Type Classification	6
4.3 Specific Purchase Timing Regression	6
4.4 Implications	6
5. Limitations	6
6. Recommendations	7
6.1 Modeling and Prediction	7
6.2 Data Context	7
6.3 Additional Options	7
7. References	8

List of Figures

Figure 1 Heatmap of Correlations between customer purchasing features.	2
Figure 2 Heatmap showing the strength of probabilities for predicting customer types.	3
Figure 3 Line graph of True versus predicted days before game for customers in test set.	5

List of Tables

Table 1 Table of validation accuracies of KNN, Logistic Regression, and Random Forest Models.....	4
Table 2 True Days Before Game and Predicted Days Before Game of ticket purchases for customers in test set.	4
Table 3 Comparison of Mean Squared errors for the given models and the new Random Forest regression model.	5

1. Introduction

The Golden State Warriors are looking to embrace data-driven marketing to enhance ticket sales and reduce email frequency. Ray Yocke's team is looking to train a machine learning model, "The Ticket Timing Model," to predict customer behavior and increase effectiveness of promotional emails. This should lead to fewer emails, higher ticket sales, and, ideally, a sold-out arena before the start of each game.

1.1 Project Scope

This case study compares multiple machine learning techniques against the current Ticket Timing Model. The dataset comprises customer demographics and purchasing behavior, including age, mailing preferences, number of tickets, seat location, ticket price, and concession purchases. The study's scope encompasses two predictive objectives: classification of fan type (Last-Minute, Planner, or In-Between) and precise estimation of purchase timing in days relative to game day.

2. Methods

2.1 Given Analysis

Initial examination of the data revealed that there is a right-skewed distribution for days between ticket purchase and game day, indicating a concentration of very last-minute purchasers. Additionally, the given correlation analysis suggested that there are features correlated with purchase timing, which helped drive the direction of this study.

The original model correctly predicted four out of five observations from the held-out test dataset, establishing a baseline for improvement.

2.2 Exploratory Data Analysis

The first steps taken in the analysis were to examine the correlations between the features in the dataset to develop an idea of relationships and possible next steps. Customer age, number of tickets purchased, seat location, and concession purchases had the strongest correlations with days before game. These most important features were selected for modeling in order to reduce unnecessary dimensionality.

2.3 Machine Learning Models for Predictions

2.3.1 Models to Predict Fan Type

The first goal of this study is to accurately classify fans by their purchasing type (Last-Minute, Planner, or In-Between). Three classification algorithms were used initially to attempt to predict fan type better than the linear models given. These models were K-Nearest Neighbors (KNN), Logistic Regression, and Random Forest. The results of these models would be compared against each other.

2.3.2 Models to Predict Number of Days

The second objective of this project was to predict the number of days a customer is likely to purchase their tickets. A Random Forest Regressor was selected for its ability to model non-linear feature interactions. Additionally, hyperparameter tuning was performed using grid search and k-fold cross-validation to optimize parameters such as tree depth, number of estimators, and minimum samples per leaf.

3. Results

3.1 Classifying Fan Type

The first step of the analysis was to create a heatmap of the correlations between the ticket purchasing data. *Figure 1* displays this heatmap. Because the focus of this study is predicting the timing of ticket purchases, it was important to note the features that are most correlated with the days before a game. In *Figure 1*, the most positively correlated features with days before game were age, number of tickets, and concession purchases. Seat location was most negatively correlated with days before a game.

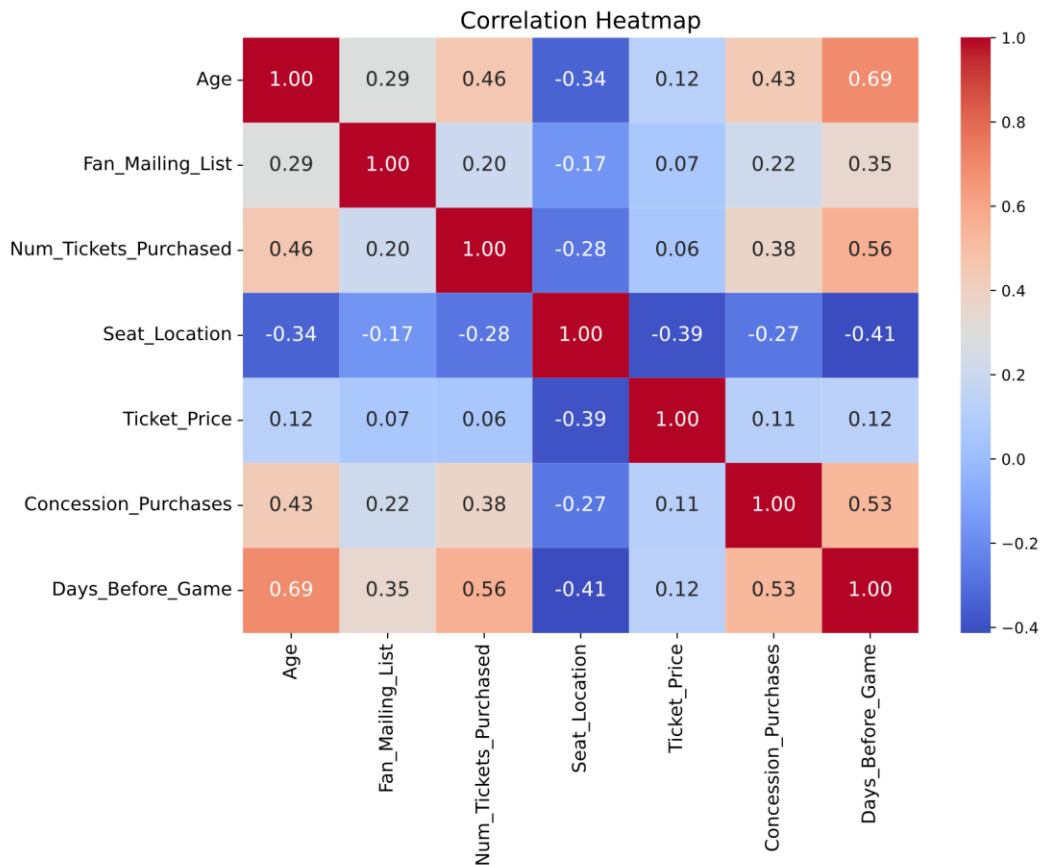


Figure 1 Heatmap of Correlations between customer purchasing features.

Case Study 2 Report

The next step of the analysis was to develop machine learning models that could potentially outperform the current models. Three machine learning models were tested: KNN, Logistic Regression, and Random Forest. Only the most correlated features were included.

All three models had 100% accuracy on the five test samples, meaning they classified all five customers into the correct customer type. Because of this, other metrics were used to decide on the best model.

Figure 2 shows the strength of the probabilities of each model when predicting each customer's type. Since all three predicted correctly, this helps represent the confidence of the models.

For another comparison, *Table 1* displays the validation accuracies of the three models.

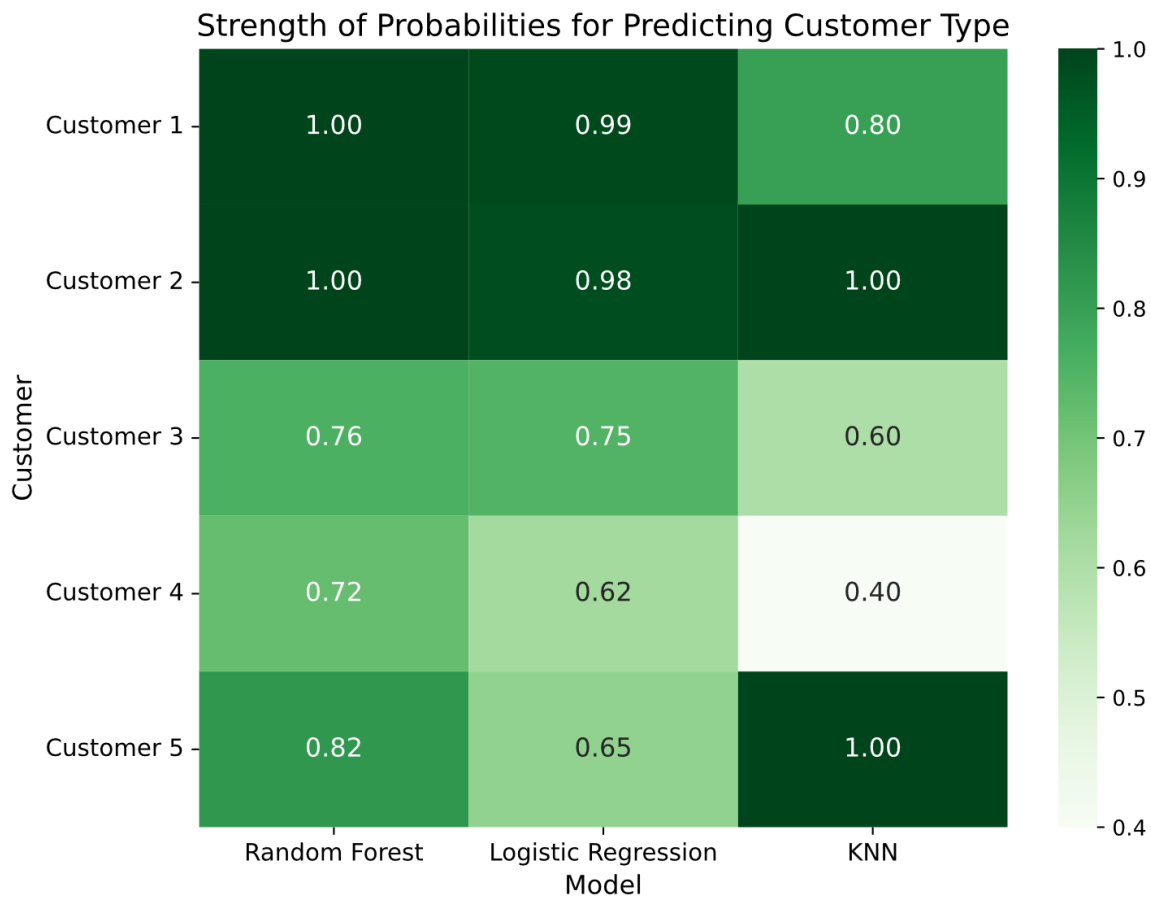


Figure 2 Heatmap showing the strength of probabilities for predicting customer types.

Table 1 Table of validation accuracies of KNN, Logistic Regression, and Random Forest Models.

	KNN	Logistic Regression	Random Forest Classification
Validation Accuracy	0.8133	0.79	0.82

3.2 Predicting Days Before Game

Table 2 presents the true number of days versus the predicted number of days before a game of the five test samples. The Random Forest Regression model closely approximated actual purchase timing. This is visually represented in *Figure 3*, which is a line graph representing the comparison of true and predicted days before game of purchases for customers in the test set.

Table 2 True Days Before Game and Predicted Days Before Game of ticket purchases for customers in test set.

True Days Before Game	Predicted Days Before Game
28	22.0
1	2.3
3	6.4
7	9.4
21	21.2

Case Study 2 Report

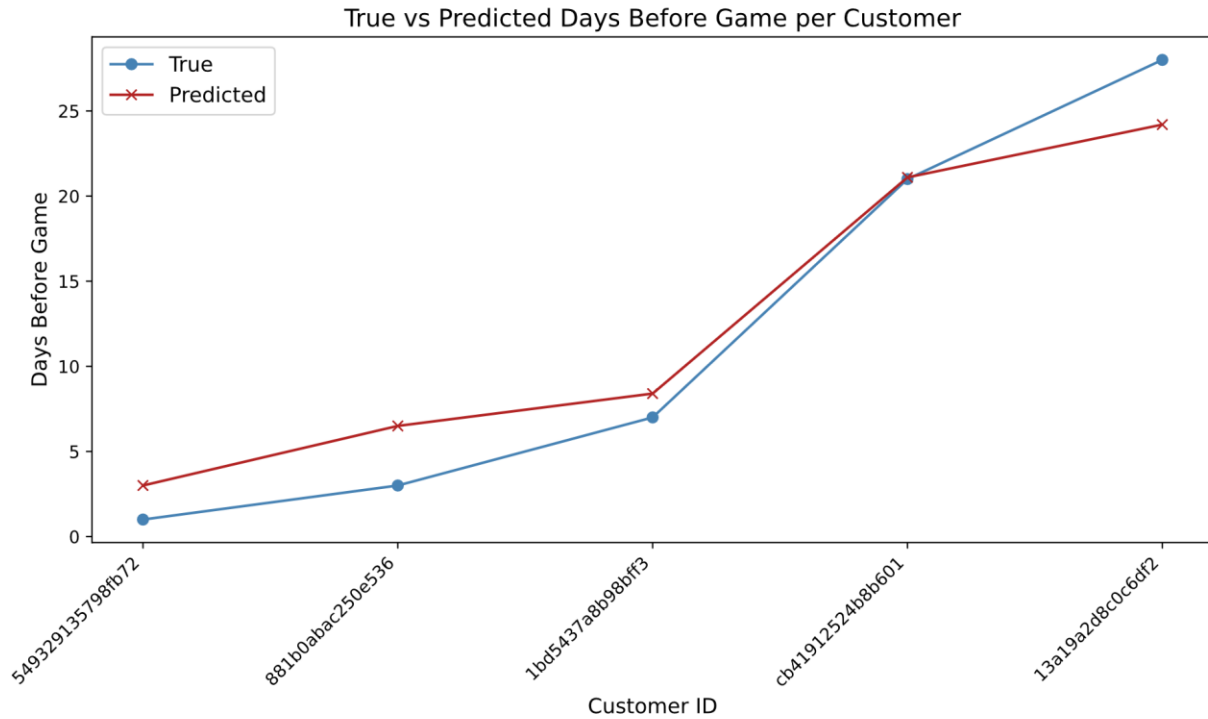


Figure 3 Line graph of True versus predicted days before game for customers in test set.

In Table 3, a comparison is shown of the mean squared errors (MSE) of the given models of simple and multiple linear regression and the new Random Forest regression model. The Random Forest regression MSE was 11.1764, significantly less than the next best given model (multiple linear regression).

Table 3 Comparison of Mean Squared errors for the given models and the new Random Forest regression model.

	Simple Linear Regression (given)	Multiple Linear Regression (given)	Random Forest Regression
Mean Squared Error	79.5456	20.0589	11.1764

4. Conclusions

4.1 Feature Relationships

The initial exploratory analysis revealed that age, number of tickets purchased, seat location, and concession purchases had the strongest correlations with the number of days before a game. Older fans and those purchasing multiple tickets were more likely to plan ahead, likely due to personal logistics. Conversely, lower seat locations correlated with last-minute purchases, possibly due to buyers waiting for late deals. These feature relationships provided a set of predictors that could be used in the models.

4.2 Fan Type Classification

Among the three classifiers tested, Random Forest attained a 0.82 validation accuracy and a perfect 1.00 test accuracy in assigning fans to Last-Minute, Planner, or In-Between. This was better than KNN (0.8133 validation), and Logistic Regression (0.79 validation). The strength of the Random Forest's probability estimations also exceeded the other models, as visualized in *Figure 2*. This ability to distinguish between groups should enable more precise email targeting windows.

4.3 Specific Purchase Timing Regression

The Random Forest Regressor with parameter tuning reduced the mean squared error to 11.18, which is a significant improvement over the previous Multiple Linear Regression's 20.06, and Simple Linear Regression's 79.55. This demonstrates the model's capacity to learn complex, non-linear dependencies among the features.

4.4 Implications

Both the classification and regression improvements translate directly to more effective email campaigns. By identifying customers' tendencies, marketing can concentrate resources depending on timing.

5. Limitations

The provided dataset had only 1,000 rows of customer information. This is quite low for machine learning models, and it likely limited model generalization to future applications.

To put this into perspective, the Chase Center that the Warriors play at has a capacity of 18,064. In relation to the capacity, and given that the average number of tickets purchased was 2.8, the data provided covered about 15.5% of that capacity of a single game. More data would be needed, as well as more context for each game, to make accurate predictions.

Additionally, certain features, such as concession purchases, may be too specific to a single game, reducing their utility for future predictions. Although concession purchases were correlated with purchase timing, and they were used in this study, it may not be that case when a more representative number of customers are included.

6. Recommendations

6.1 Modeling and Prediction

Implementation of the Random Forest Classifier for fan segmentation and the Random Forest Regressor for specific purchase-timing prediction is advised. Logistic Regression performed similarly and could be used as a secondary option. The predicted category of fans should be used for future promotional emailing.

6.2 Data Context

To enhance predictive power, the Golden State Warriors should expand data collection to include more games and more customers, and possibly pre-purchase indicators such as webpage visits. It also might be useful to include details specific to games, such as day of the week and opponent.

6.3 Additional Options

According to the given data, about 29% of buyers purchase their tickets within two days of gameday. With this time period's high frequency of purchases, using hyper-targeted, time-sensitive campaigns could capture this last-minute demand more effectively and smoothly. These strategies could include real-time inventory alerts, countdown messaging, or pricing incentives. This would capitalize on the high volume of fans making their decision in the last few days before a game.

7. References

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. <https://scikit-learn.org>.

Statista. (2025). Gate receipts of the Golden State Warriors (NBA) from 2015 to 2024. Retrieved from <https://www.statista.com/statistics/381541/gate-receipts-of-the-golden-state-warriors-national-basketball-association/>.